ARTICLE | Article in Press

Data-driven insights into stainless steel pitting corrosion through large-scale literature mining and database construction

Zhen Huang · Jinyan Duan · Yupeng Liu · Changgang Wang

Received: 10 July 2026 / Accepted: 6 October 2026
© The Author(s) 2026

Abstract

Pitting corrosion is one of the most prevalent and detrimental forms of localized corrosion in stainless steels exposed to chloride-containing environments. Its initiation and propagation are governed by the combined effects of alloy composition, environmental conditions, and experimental parameters. Although a substantial body of experimental data has been accumulated through decades of research, these data remain largely dispersed across the scientific literature, limiting their accessibility for large-scale statistical analyses and data-driven investigations. To address this challenge, this study proposes a large language model (LLM)-based framework for structured data extraction and database construction from scientific literature. By decomposing the complex information extraction process into multiple subtasks, including experimental condition extraction, pitting potential extraction, and data alignment, the framework enables the automated acquisition and integration of experimental data from unstructured scientific documents. Using the proposed framework, 12,792 candidate publications retrieved using a broad stainless-steel pitting corrosion search strategy were processed, resulting in a structured extraction output containing 36,095 experimental records. The database encompasses key information, including alloy compositions, environmental parameters, and pitting potentials. Stainless-steel-specific descriptive and statistical analyses were subsequently conducted using material-screened subsets of the extracted records. Data quality evaluation demonstrates that the proposed framework achieves strong aggregate performance in structured information extraction from complex scientific texts, while stage-specific validation indicates that pitting-potential extraction and normalization remain more challenging than the overall field-level extraction task. Statistical analysis of the resulting database reveals that existing studies are predominantly focused on room-temperature, neutral chloride environments, as well as austenitic and duplex stainless steel systems. Furthermore, partial correlation analysis revealed conditional associations between the investigated variables and pitting potential. Among these variables, molybdenum exhibited the strongest positive conditional association, whereas temperature showed a negative conditional association. Chloride ion concentration showed only a weak negative association. These trends were generally consistent with established corrosion knowledge and therefore provided a scientific plausibility check for the statistical patterns observed in the constructed database. This work highlights the potential of large language models for knowledge extraction and automated database construction from corrosion-related scientific literature. The proposed framework

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.



https://doi.org/10.1038/s41598-026-75356-9

Published online: 19 October 2026

provides a reusable data resource and technical pathway for corrosion informatics research, while establishing a foundation for future data-driven studies on corrosion mechanism discovery, performance prediction, and materials design.

Keywords Stainless steel · Pitting corrosion · Large language models · Scientific literature mining · Corrosion informatics · Structured database construction

Introduction

Stainless steels are widely utilized in critical sectors, including marine engineering, energy infrastructure, chemical processing facilities, transportation systems, and nuclear power plants, owing to their excellent mechanical properties, manufacturability, and corrosion resistance[1–3]. In chloride-containing environments, a passive film formed on the stainless steel surface effectively isolates the metallic substrate from corrosive media, thereby providing the material with superior corrosion resistance. However, when the passive film is locally disrupted by chloride ion attack, highly localized corrosion pits may initiate on the material surface. These pits can subsequently promote the development of more severe degradation processes, such as crevice corrosion, stress corrosion cracking, and corrosion fatigue[4–6]. Due to its concealed nature, rapid propagation, and potentially catastrophic consequences, pitting corrosion is widely regarded as one of the most detrimental forms of localized corrosion affecting stainless steels during service[7,8].

Numerous studies have demonstrated that the pitting corrosion behavior of stainless steels is governed by the combined effects of alloy composition, microstructural characteristics, and environmental conditions. Among these factors, alloying elements such as chromium (Cr), molybdenum (Mo), and nitrogen (N) are widely recognized as key contributors to enhanced pitting corrosion resistance and form the basis of the classical Pitting Resistance Equivalent Number (PREN) model[9–11]. Meanwhile, environmental parameters, including temperature, chloride ion concentration, pH, scan rate, and electrolyte composition, have also been shown to significantly influence passive film stability and localized corrosion processes[12–14]. For example, increasing temperature generally reduces passive film stability and increases susceptibility to pitting corrosion, whereas elevated chloride ion concentrations accelerate localized passive film breakdown and promote pit initiation[15–17]. Although extensive experimental investigations have been conducted on individual stainless steel grades or under specific environmental conditions, substantial variations in experimental conditions, testing methodologies, and material systems often limit the consistency and comparability of reported findings. Consequently, extracting generalizable trends from the vast body of dispersed literature data and establishing a statistical understanding that spans different material systems and environmental conditions have become important research objectives in the field of corrosion science.

In recent years, the rapid development of Materials Informatics has provided new opportunities for investigating complex material behaviors[18,19]. By leveraging high-throughput computation, big data analytics, and machine learning techniques, researchers can uncover underlying patterns from large volumes of experimental and computational data, thereby enabling materials property prediction and design optimization[20]. However, compared with fields such as battery materials, catalytic materials, and structural materials, the corrosion domain remains relatively data-deficient. This limitation primarily arises from the fact that corrosion-related experimental data are highly fragmented across the scientific literature, with a substantial proportion of critical information embedded in textual descriptions, tables, and figures. As a result, such information cannot be readily transformed into structured datasets suitable for statistical analysis and machine learning applications. This data acquisition bottleneck has significantly hindered the development of Corrosion Informatics.

The advancement of Natural Language Processing (NLP) technologies has provided new opportunities for the automated extraction of information from scientific literature. Traditional approaches typically rely on keyword matching, rule-based extraction, or supervised learning models to perform data extraction tasks. However, these

methods often exhibit limited performance when dealing with complex table parsing, multi-variable relationship identification, and cross-paragraph information integration[21]. In recent years, Large Language Models (LLMs), exemplified by the GPT series, have demonstrated remarkable capabilities in text comprehension, information extraction, and complex reasoning tasks[22–24]. Through their ability to capture and interpret contextual semantics, LLMs can automatically identify critical information from unstructured scientific documents, including experimental conditions, material compositions, and performance metrics, thereby substantially reducing the cost and effort associated with database construction[25–29]. At the same time, considerable research efforts have been devoted to addressing challenges associated with LLMs, including hallucinations, information loss in long-context processing, and the stability of structured outputs. Various optimization strategies and constraint mechanisms have been proposed to mitigate these issues[30–32], thereby providing a more reliable technical foundation for the automated extraction of scientific data.

Although LLMs have demonstrated considerable potential in fields such as biomedicine, chemistry, and materials science, their application in corrosion science remains at an early stage. To date, most studies on stainless steel pitting corrosion have been conducted as individual experimental investigations, with relatively limited efforts devoted to large-scale analyses based on literature-derived datasets. In particular, current understanding of the relationships among alloy composition, environmental conditions, and pitting potential is largely derived from limited experimental datasets and empirical models, while comprehensive statistical validation across multiple studies, stainless steel grades, and environmental conditions remains lacking. Furthermore, substantial variations in experimental methodologies, testing conditions, and data reporting practices exist among different studies, making traditional manual data curation approaches inadequate for large-scale database construction. Therefore, the development of a high-quality database for stainless steel pitting corrosion, together with systematic statistical analyses based on such a database, is important for advancing data-driven corrosion research.

Motivated by these challenges, this study proposes an LLM-assisted data extraction framework for corrosion science literature. By combining task decomposition with a structured output strategy, the framework enables the automated extraction and integration of pitting-corrosion-related information from a broad literature corpus. Using the proposed framework, 12,792 candidate publications retrieved using a broad stainless-steel pitting corrosion search strategy were processed, resulting in a large-scale structured extraction output containing 36,095 experimental records. Material-screened subsets were subsequently used for stainless-steel-specific descriptive and statistical analyses. The database encompasses key variables, including alloy compositions, experimental environmental parameters, and pitting potentials. Building upon this database, data quality assessment, statistical profiling, and partial correlation analysis were further performed to systematically investigate the conditional associations between alloy composition, environmental factors, and pitting potential. The present work not only provides a reusable data resource and technical framework for corrosion informatics research but also demonstrates a practical application of Large Language Models for knowledge extraction from materials science literature.

Method

In this study, a prompt-based Large Language Model (LLM) framework was employed to extract structured experimental data from scientific literature (Fig. 1). Unlike conventional approaches that require training task-specific natural language processing models, the proposed framework leverages carefully designed prompts to guide the LLM during the information extraction process. These prompts explicitly define the extraction objectives, data schema, and output constraints. By incorporating domain-specific instructions into the prompts, the model is able to interpret the content of scientific publications and transform unstructured experimental descriptions into structured records.

Prompting strategies can generally be categorized into three types:

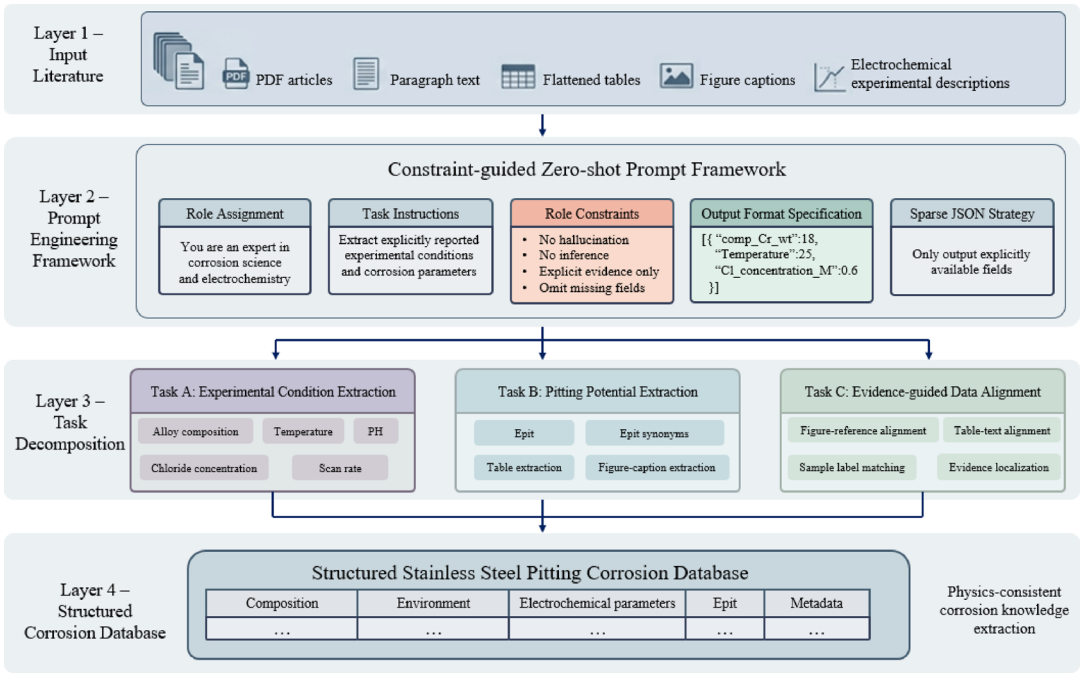


Fig. 1 Architecture of the prompt-based scientific information extraction framework.

(1) Zero-shot prompting[33,34], in which only task instructions are provided in the prompt, without any example inputs or outputs;

(2) Few-shot prompting[35], in which several task-specific examples are included in the prompt to guide the model toward generating more consistent and standardized outputs;

(3) Chain-of-thought prompting[36], which encourages the model to perform intermediate reasoning based on the input before generating the final response, thereby improving its ability to satisfy complex task requirements.

In the present study, a zero-shot prompting strategy was adopted. As illustrated in Fig. 2, the designed prompt consists of three main components:

(1) Task instructions, which define the objectives of the information extraction task;

(2) Rule constraints, which specify the conditions and restrictions governing the extraction process;

(3) Output format specifications, which constrain the model to generate structured outputs suitable for subsequent automated processing.

LLMs rely on attention mechanisms to dynamically focus on relevant portions of the input text. However, when processing lengthy scientific documents containing diverse types of information, the model's attention may become dispersed across irrelevant content, thereby adversely affecting the accuracy of information extraction[37,38]. The fundamental principle of the attention mechanism can be expressed as follows:

$$Attention(Q,K,V) = soft\max\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where Q, K, and V denote the query, key, and value matrices, respectively; K^T represents the transpose of the key matrix; and d_k denotes the dimensionality of the key vectors. The softmax function normalizes the scaled query–key similarity scores into attention weights.

To mitigate this issue, the information extraction process was decomposed into multiple subtasks rather than attempting to extract all information through a single prompt. The overall workflow consists of four major stages:

(1) Document preparation and preprocessing;

(2) Prompt-based extraction of experimental conditions;

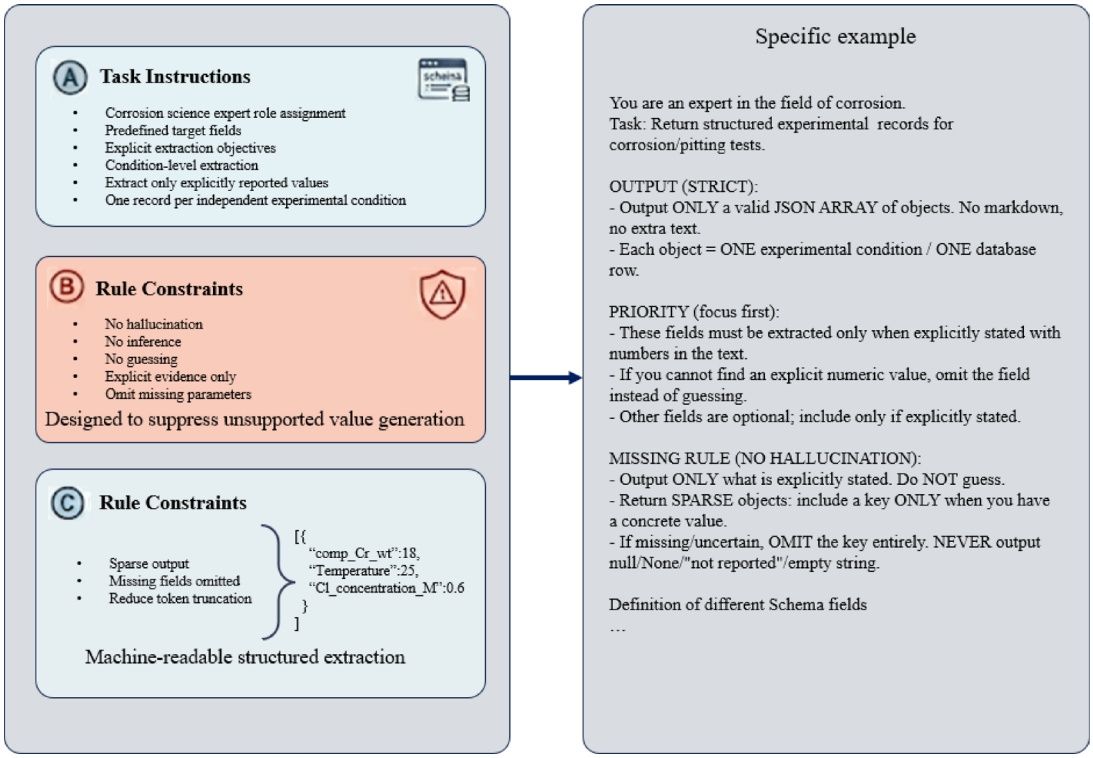


Fig. 2 Template of constraint-guided Zero-shot Prompt.

(3) Pitting potential extraction;

(4) Data alignment and structured database construction.

This task decomposition strategy helps direct the model's attention toward information that is most relevant to the current task, thereby improving the reliability and stability of the extraction results. Furthermore, decomposing the overall workflow into smaller subtasks effectively reduces the number of output tokens generated in each model invocation, thereby alleviating issues associated with token limitations and incomplete responses[39,40].

Considering that this study involves large-scale automated information extraction from more than 12,000 corrosion-related publications, the LLM configuration was selected by jointly considering scientific literature comprehension, structured information extraction capability, extraction accuracy, reliability, inference efficiency, and deployment cost. To quantitatively evaluate the model selection, a comparative benchmark was conducted using five model configurations: GPT-3.5, Gemini Flash, DeepSeek-v3.2, Qwen-Max, and the proposed DeepSeek-v3.2–Qwen-Max hybrid configuration.

To ensure a fair comparison, all model configurations were evaluated using the same preprocessing procedure, task-decomposition workflow, prompt instructions, output schema, and a representative validation subset of 100 publications. Model performance was assessed in terms of precision, recall, F1 score, hallucination rate, valid-JSON output rate, processing failure rate, average processing time, and approximate processing cost. Condition–result correspondence was additionally examined as part of the record-level correctness assessment. The quantitative benchmark results are presented in Sect. "Results and discussion".

Based on the comparative evaluation, DeepSeek-v3.2 and Qwen-Max were adopted in a task-specific hybrid configuration for large-scale database construction. DeepSeek-v3.2 was employed for experimental-condition extraction (Task A), whereas Qwen-Max was used for pitting-potential extraction (Task B) and condition–result alignment (Task C). This task-specific allocation was designed to exploit the complementary capabilities of the two models while maintaining a practical balance between extraction performance and computational cost.

Document preparation and preprocessing

The source literature corpus used in this study was retrieved from the ScienceDirect database using the keyword query "stainless steel pitting corrosion". The publication period was restricted to 2019–2026, and the document type was restricted to research articles. Literature retrieval was completed on April 23, 2026. Based on these search criteria, 12,792 publications were retrieved and acquired in Portable Document Format (PDF), and all retrieved publications were directly included in the subsequent automated information-extraction workflow without additional manual material-level screening or deduplication.

The literature retrieval stage was intentionally designed as a broad, high-recall search to minimize the risk of excluding potentially relevant studies. Because the ScienceDirect keyword search is relevance-based rather than an exact material-domain classification, the retrieved corpus could also include publications involving non-stainless materials. Accordingly, the 12,792 publications represent the candidate literature corpus used for automated information extraction rather than a manually verified set of stainless-steel-only publications.

All publications included in the corpus were peer-reviewed articles published in SCI-indexed journals, providing a baseline level of quality control for the extracted experimental data. Although journal-level impact and ranking may vary, impact factors or journal rankings were not used to weight individual records because these bibliometric indicators do not directly reflect the reliability of individual experimental measurements. Therefore, all extracted records were treated equally in the subsequent analysis.

To further characterize the application coverage of the literature corpus, the retrieved publications were classified into broad application-oriented research fields based on application-related keywords identified from the title and first-page textual content. As shown in Fig. 3, the corpus covers a diverse range of fields, including manufacturing and surface engineering, nuclear and energy, marine and offshore applications, biomedical applications, construction and infrastructure, and chemical and process industries, as well as general corrosion and materials research. The uneven distribution across these fields reflects differences in research attention among application areas. Accordingly, the statistical trends derived from the complete corpus should be interpreted as overall cross-literature patterns rather than as relationships that are necessarily equally generalizable to every specific application field. The preprocessing workflow began with the parsing of PDF documents to extract experimental information presented in both narrative text and tabular formats. The PyMuPDF library was employed to extract page-level textual content, converting the original PDF layouts into machine-readable text sequences.

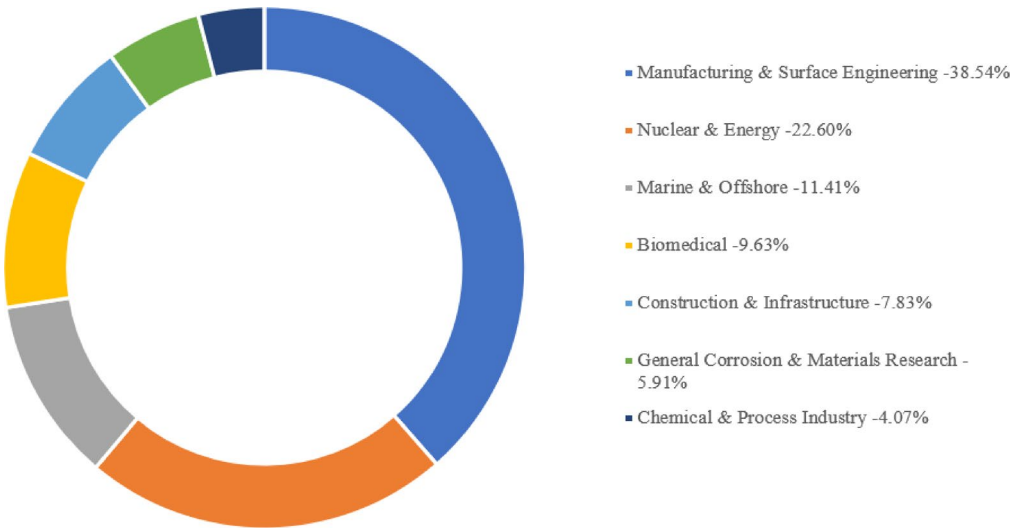


Fig. 3 Distribution of the literature corpus across major application fields.

In addition to narrative text, experimental conditions in corrosion-related literature are frequently reported in tabular form. Therefore, the pdfplumber library was further employed to extract table data from the PDF documents. This tool is capable of identifying table structures and retrieving cell-level information. To facilitate subsequent processing by Large Language Models (LLMs), each extracted table was converted into a linearized textual representation that preserved both the column headers and the row-wise data entries. Under this representation, each table row was treated as an individual experimental condition, enabling its direct mapping to a structured database record.

Subsequently, the extracted narrative text and the reconstructed table text blocks were merged into a unified textual input, enabling the Large Language Model (LLM) to process heterogeneous information sources, including descriptive text and tabular data, in a consistent manner. This unified representation served as the input for the subsequent prompt-based information extraction workflow. The overall preprocessing and data integration procedure is illustrated in Fig. 4.

Prompt-based experimental condition extraction (Task A)

Experimental conditions were extracted using a prompt-based LLM framework. A zero-shot prompting strategy was adopted, whereby the model was guided to perform information extraction solely through carefully designed natural language instructions without the provision of task-specific examples. To ensure the accuracy and reliability of the extracted results, the prompts explicitly encoded the extraction objectives, schema constraints, and extraction rules. The prompt design consisted of three main components: task instructions, rule constraints, and output format specifications. This design enabled the LLM to perform controlled information extraction from unstructured scientific texts.

Task Instructions: In this study, the LLM was instructed to perform the information extraction task while assuming the role of a domain expert in corrosion science[41]. A predefined set of target fields was specified in the prompt (Fig. 5), corresponding to the schema of the structured database. Each field was accompanied by an explicit definition, and the model was instructed to extract a value only when the corresponding information was explicitly

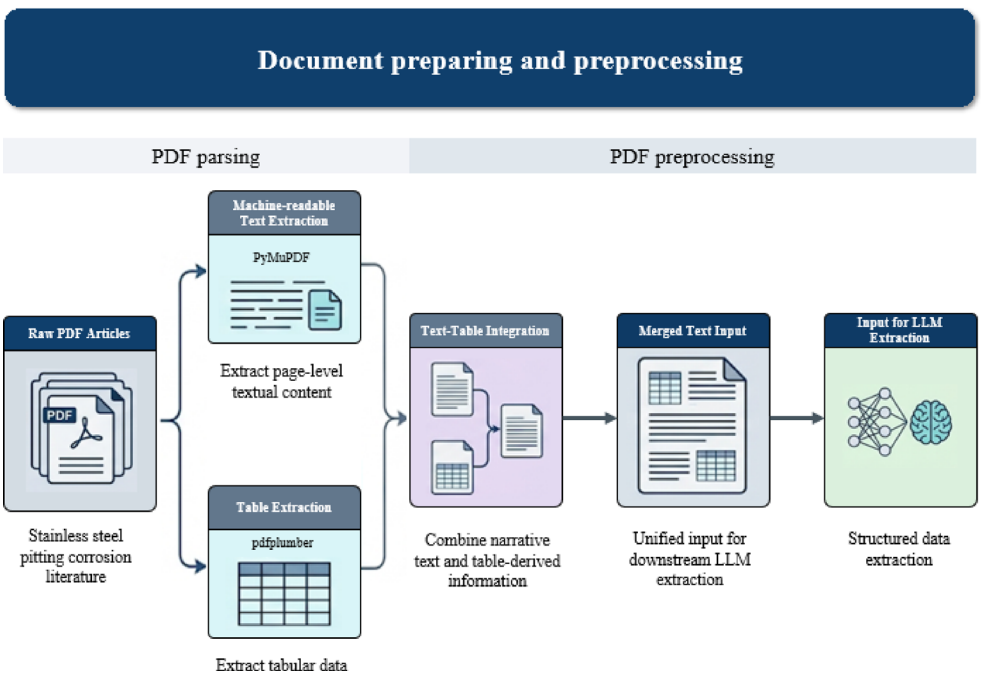


Fig. 4 Document preparing and preprocessing.

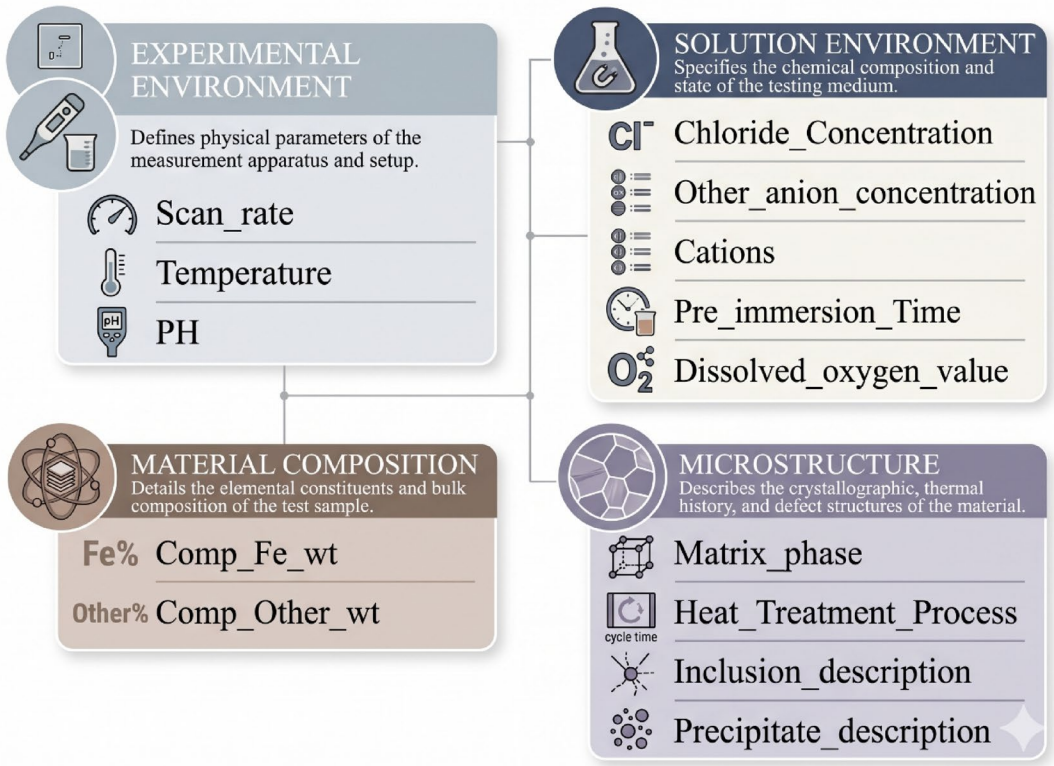


Fig. 5 Target extraction field.

reported in the source text. This constraint ensured that all extracted data remained directly traceable to the original literature and minimized potential biases arising from implicit inference.

All extracted information was mapped to a predefined set of output fields to ensure consistency across different publications. In addition, an experimental condition-level extraction strategy was adopted in this study. When multiple experimental conditions were reported within a single publication, such as variations in alloy composition, solution chemistry, or testing parameters, the model was instructed to generate multiple independent records. Each record corresponded to a distinct experimental condition, thereby preserving the one-to-one correspondence between experimental conditions and extracted data entries. The overall extraction workflow is illustrated in Fig. 6.

Rule Constraints: Model hallucination refers to the tendency of LLMs to generate information that is not explicitly present in the input text but appears contextually plausible or coherent[42]. In information extraction tasks, hallucinations often manifest as the inference of missing numerical values or the generation of details that are not explicitly reported, thereby significantly compromising the reliability of structured datasets.

To mitigate this issue, a set of stringent rule constraints was incorporated into the prompt design[43]. The model was explicitly instructed to extract only information that was explicitly reported in the source text and to avoid any form of inference, estimation, or speculation. For parameters that were not reported in the original document, the corresponding fields were omitted rather than populated with placeholder values. This constraint-based design ensured that all extracted data were evidence-based and directly traceable to the source literature. Consequently, the reliability, consistency, and transparency of the resulting dataset were substantially improved.

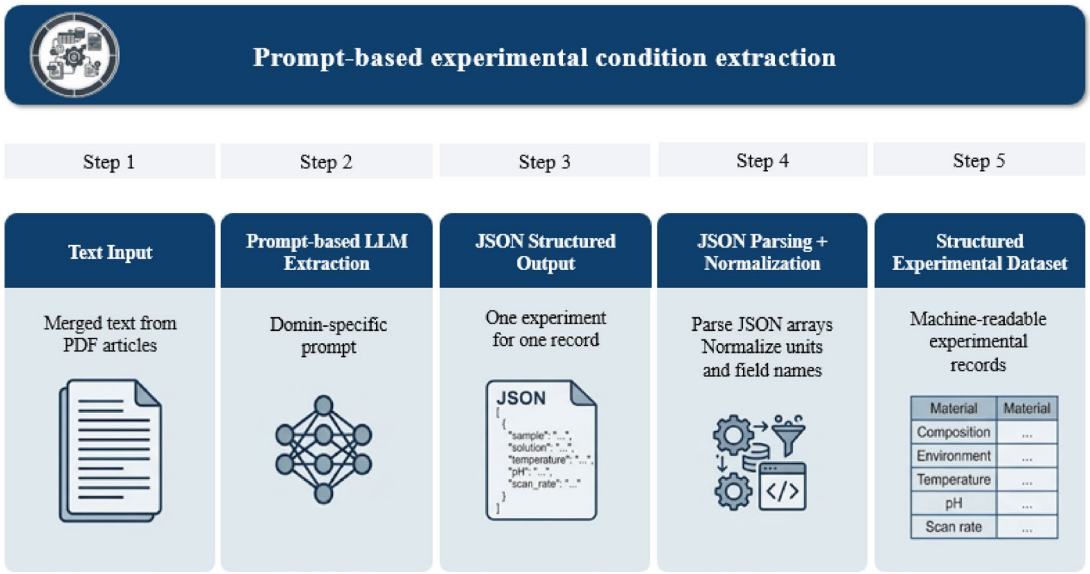


Fig. 6 Experiment condition extraction process.

Output Format Specifications: The extraction output was defined as a JavaScript Object Notation (JSON) array, in which each object represented an independent experimental condition. This design enabled multiple experimental settings reported within a single publication to be represented as separate structured records.

To mitigate response truncation and incomplete outputs caused by excessive input–output token lengths, a sparse output strategy was adopted. Under this strategy, each JSON object contained only those fields that were explicitly reported in the source document, while fields corresponding to missing parameters were omitted. This approach effectively reduced the number of output tokens and improved the robustness and consistency of the generated results[44].

Subsequently, the generated JSON array was parsed into individual database records, and the extracted parameters were standardized through unit normalization to ensure consistency across different data sources. Finally, all processed records were integrated to construct the structured database of experimental conditions.

Pitting potential extraction (Task B)

In this study, the target parameter was the pitting potential (E_{pit}). The pitting potential, also referred to as the critical pitting potential, is a key electrochemical parameter used to evaluate the susceptibility of metals to localized corrosion. It is defined as the potential at which the passive film undergoes stable breakdown, resulting in the initiation of pitting corrosion. In potentiodynamic polarization tests, pitting potential is typically identified as the potential corresponding to a pronounced and sustained increase in anodic current density, indicating the onset of pit formation. In general, a higher pitting potential value corresponds to greater resistance to pitting corrosion. Consequently, this parameter is widely used as an important indicator for comparing the corrosion performance of different alloys under various environmental conditions.

In this subtask, a high-recall candidate generation procedure was employed to identify potentially relevant text segments, and the overall workflow is illustrated in Fig. 7. Candidate blocks were extracted from both narrative text and tabular data. For paragraph-level candidates, filtering was performed based on the co-occurrence of corrosion-related trigger terms and potential-value patterns. For table-level candidates, relevant content was identified using structural indicators (e.g., “Table X”) together with keyword-based filtering. This rule-based strategy ensured that all potentially relevant regions were retained for subsequent analysis.

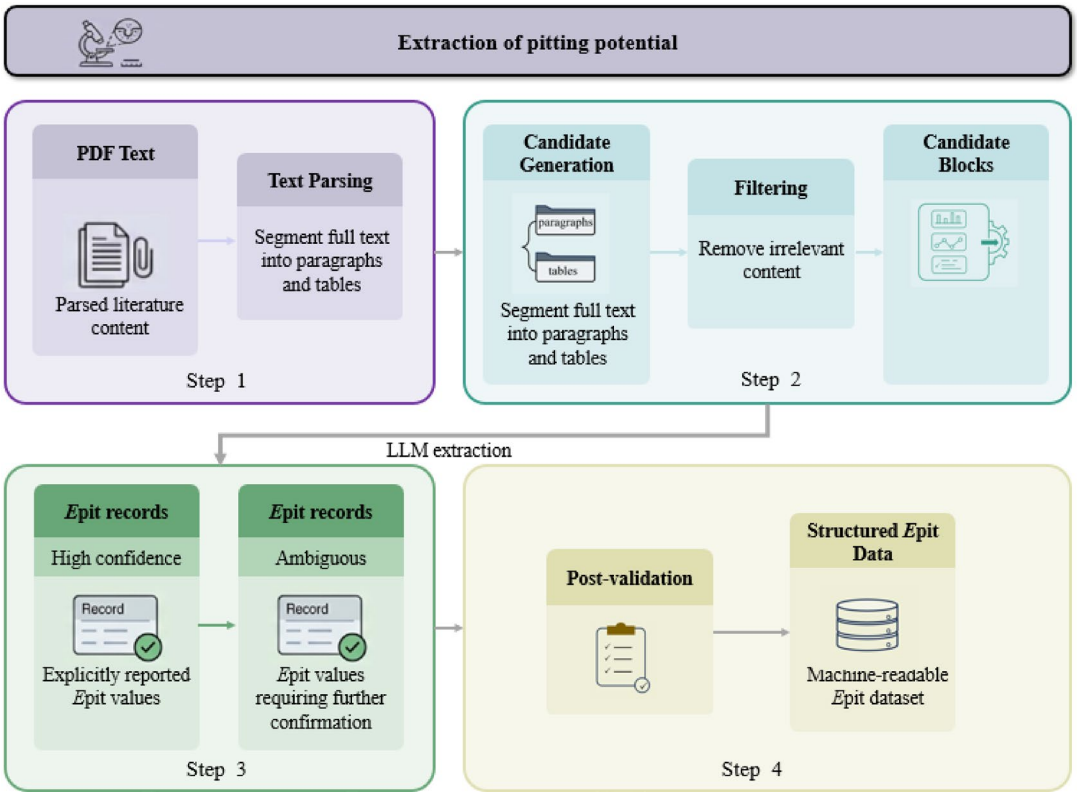


Fig. 7 Extraction process of pitting potential.

Subsequently, the filtered candidate blocks were processed using an LLM-based extraction module. The model was instructed to identify and extract pitting-related potential values that were explicitly reported in the source text and to return the results in a structured JSON format. To address ambiguity in terminology and reporting conventions, the extracted results were categorized into two groups: epit_records, representing high-confidence pitting potential values, and epit_candidates, representing potential candidates with weaker textual evidence or whose interpretation depended on the surrounding context. These candidate entries were subjected to further evaluation in subsequent processing stages.

To ensure data reliability, an evidence-based information extraction strategy was adopted. Each extracted value was required to be supported by a corresponding evidence text (*evidence_text*), which contained both the relevant terminology and the associated numerical value. The model was explicitly instructed to avoid any form of inference, estimation, or speculation, thereby ensuring that all extracted data remained directly traceable to the original literature.

Finally, the extracted results underwent a post-processing stage for validation and standardization. This procedure included the removal of invalid entries, standardization of numerical representations and units, and identification of the reference electrodes associated with the reported pitting potentials, including SCE, Ag/AgCl, SHE, and RHE. To improve comparability across studies, pitting-potential values were subsequently harmonized to the saturated calomel electrode (SCE) scale. For records in which the reference electrode was not retained in the structured extraction output, the corresponding source publications were re-examined to recover the reference-electrode information; these records were confirmed to have been measured versus SCE and were therefore treated as SCE-referenced potentials. Potentials reported against other identifiable reference electrodes were converted to the SCE scale using the corresponding reference-electrode potentials. Records for which a reliable conversion could not be established were excluded from analyses requiring harmonized pitting-potential values.

Data alignment and structured database construction (Task C)

In corrosion studies, a single publication often investigates multiple experimental conditions and consequently reports multiple corresponding experimental outcomes. Therefore, accurately aligning these results with their associated experimental conditions is a critical step in the construction of a reliable structured database.

Traditional approaches based on natural language processing or deep learning typically require large volumes of annotated data or rely on complex feature engineering to capture the relationships between experimental conditions and experimental outcomes. In contrast, LLMs possess strong contextual understanding and reasoning capabilities, enabling them to infer the associations between experimental conditions and their corresponding results directly from the original text. These capabilities make LLMs particularly well suited for condition–outcome alignment tasks in scientific literature.

In this study, an evidence-guided LLM-based alignment strategy was developed to associate the extracted experimental condition records with their corresponding pitting potential values. The alignment framework consisted of the following components, as shown in Fig. 8:

Alignment inputs and evidence construction

The alignment process integrated three categories of inputs: (1). Structured experimental condition records extracted in Task A; (2). Structured pitting potential records extracted in Task B; (3). Page-level textual evidence retrieved from the original PDF documents.

To reduce irrelevant contextual information and improve alignment efficiency, only pages located in the vicinity of the extracted pitting potential entries were selected as supporting evidence. This localized evidence retrieval strategy helped focus the model's attention on relevant content while controlling the input token length.

Constraint-guided alignment using large language models

A constraint-guided alignment strategy based on Large Language Models (LLMs) was employed in this study. For each experimental condition record, the model was permitted to select at most one corresponding pitting potential record from the outputs generated in Task B. If no reliable match could be established, a null value was returned.

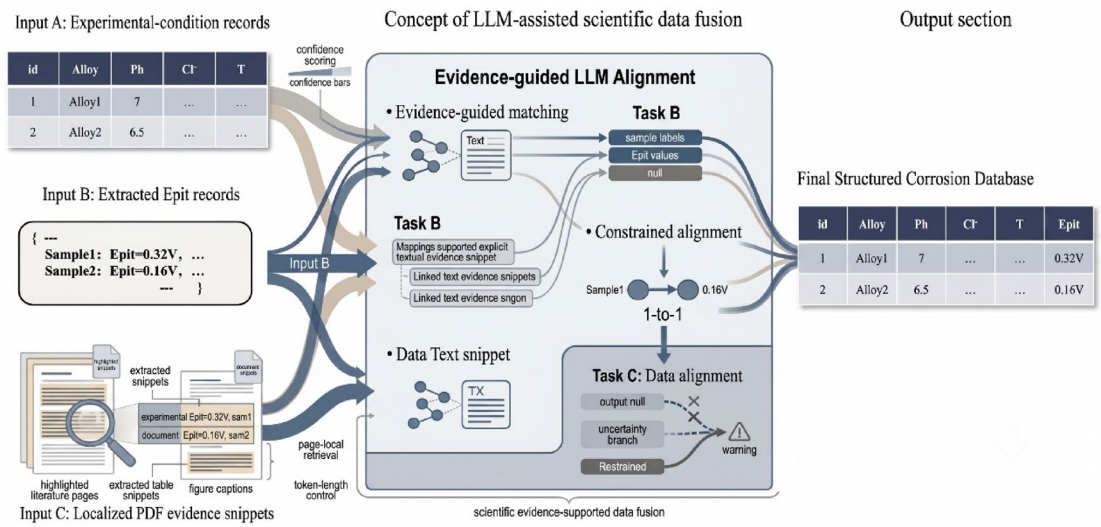


Fig. 8 Evidence-guided LLM constrained alignment and structured database construction.

The alignment process was performed in an evidence-guided manner, requiring the model to rely on explicit textual evidence to justify each mapping relationship. In particular, when pitting potential values were reported in tabular form, the model was required to reconstruct the table structure from the flattened PDF text, identify the relevant columns (e.g., pitting potential columns), and establish the mapping between sample identifiers and the reported numerical values.

When explicit identifiers, such as sample labels, were available, they were used as the primary matching criteria. In the absence of such identifiers, secondary attributes were utilized to infer the corresponding relationships, including alloy composition, solution chemistry, temperature, scan rate, and heat-treatment conditions.

To ensure the reliability of the alignment results, the model was restricted to selecting from existing pitting potential records and was explicitly prohibited from generating new numerical values. When multiple candidate matches existed and insufficient evidence was available to support a unique mapping, the alignment result was conservatively assigned as null. This strategy effectively reduced the risk of erroneous associations and improved the overall reliability of the constructed database.

Post-processing and database integration

The alignment outputs were generated in a structured format, including the index of the selected pitting potential record and the confidence level associated with each mapping. Subsequently, the aligned pitting potential values, together with their corresponding units and metadata, were merged back into the experimental condition records obtained from Task A.

Finally, all successfully aligned records were integrated to construct the complete structured database, in which each record contained both the experimental condition information and its corresponding corrosion performance parameter (E_{pit}). Through this process, heterogeneous information extracted from scientific literature was transformed into a unified, machine-readable dataset.

Validation and statistical evaluation

To provide a more comprehensive assessment of the reliability of the proposed extraction framework, the manual validation sample was expanded to 300 publications. The original 100-publication comparative benchmark constituted a subset of this expanded validation sample. Among these, 236 publications contained eligible experimental information for the stratified characterization summarized in Table 1. Each publication was manually reviewed against the original article to establish the ground-truth extraction records. Because the literature corpus was retrieved using keyword-based searches, not all retrieved publications necessarily contained experimental records satisfying the predefined extraction criteria. Publications without eligible target data were retained as negative cases during validation, for which an empty extraction was considered the correct outcome rather than a processing failure. The validation set was further characterized according to experimental complexity, pitting-potential availability, and alloy family to assess extraction performance across heterogeneous experimental and materials-science contexts.

Table 1 Composition and characteristics of the manually validated publication set.

Validation dimension	Category	Number of publications (*N*)	Proportion (%)	Total (*N*)
Experimental complexity	Single-condition	109	46.19%	236
	Multi-condition	127	53.81%	
Pitting-potential availability	With E_{pit}	63	26.70%	236
	Without E_{pit}	173	73.30%	
Alloy family	Aluminum alloys	62	26.27%	236
	Stainless steels	98	41.53%	
	Carbon/low-alloy steels	41	17.37%	
	Other / Unclassified	35	14.83%	

The material-family distribution of the eligible publications also reflects the intentionally broad retrieval strategy described in Sect. "Document preparation and preprocessing". Although the search query was centered on "stainless steel pitting corrosion", material identity was not used as an inclusion criterion during the initial literature retrieval. Consequently, some retrieved publications contained eligible pitting-corrosion experiments involving non-stainless materials, including aluminum alloys and carbon/low-alloy steels. These publications were retained in the validation sample because the validation was intended to evaluate the extraction framework on the actual candidate corpus. Thus, the non-stainless publications in Table 1 represent valid extraction cases rather than negative retrieval cases. Material-domain filtering was subsequently applied at the record level for stainless-steel-specific downstream analyses.

Extraction performance was evaluated at the value level using precision, recall, and F1 score. True positives (TP) were defined as extracted values that correctly matched the manually established ground-truth values, false positives (FP) as extracted values that were incorrect or unsupported by the source publication, and false negatives (FN) as ground-truth values that were not successfully extracted. Accordingly, precision, recall, and F1 score were calculated as follows:

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$F1 = \frac{2 \times Precision \times \mathrm{Re}call}{Precision + \mathrm{Re}call} \tag{4}$$

To quantify the statistical uncertainty of the performance estimates while accounting for the non-independence of multiple records extracted from the same publication, 95% confidence intervals (CIs) for precision, recall, and F1 score were estimated using publication-level bootstrap resampling. In each bootstrap iteration, publications were sampled with replacement, and all records associated with each selected publication were retained as a cluster. Precision, recall, and F1 score were recalculated for each resampled dataset. The 2.5th and 97.5th percentiles of the resulting bootstrap distributions were used as the lower and upper bounds of the 95% CIs.

To provide a more diagnostic assessment of extraction reliability beyond conventional precision, recall, and F1 score, extraction errors were further classified into eight mutually exclusive categories according to their primary error mechanism: (1) hallucinated value, (2) wrong field assignment, (3) incorrect unit, (4) incorrect reference-electrode representation, (5) table row or column misalignment, (6) incorrect condition–result association, (7) duplicated record, and (8) missed value. Each identified error was assigned to a single category according to a predefined priority-based classification scheme to avoid double counting.

The error counts used in this diagnostic taxonomy are distinct from the value-level FP and FN events used to calculate precision and recall. A single underlying extraction error may contribute to both an FP and an FN in the value-level evaluation. For example, when an incorrect value is extracted in place of the corresponding ground-truth value, the incorrect prediction contributes an FP, while the unmatched ground-truth value contributes an FN. In contrast, the same underlying error is counted only once in the diagnostic error taxonomy and is assigned to a single primary error category. Consequently, the sum of FP and FN events is not expected to equal the total number of errors reported in the error-attribution analysis.

Accordingly, the "missed value" category in the diagnostic taxonomy represents omission-type errors and is not numerically equivalent to the total number of FN events. FN events may also arise when a value is extracted but fails to match the corresponding ground-truth value because of an incorrect extraction, representation, or association (see Table 2).

Table 2 Definition and classification criteria for extraction errors.

Error category	Definition
Hallucinated value	An extracted value that could not be verified anywhere in the corresponding source publication or manually established ground truth
Wrong field assignment	A value explicitly reported in the source and present in the ground truth but assigned to an incorrect parameter field (e.g., Ecorr assigned as E_{pit})
Incorrect unit	The extracted numerical value and corresponding experimental condition were correct, but the associated unit was incorrect
Incorrect reference-electrode representation	The source potential was identified, but the converted potential value or reference-electrode assignment was incorrect
Table row or column misalignment	A group of extracted fields was systematically associated with an incorrect table row or column
Incorrect condition–result association	The extracted parameter and numerical value were correct but were associated with an incorrect alloy, specimen, or experimental condition
Duplicated record	A record identical to an existing correctly extracted record was generated more than once
Missed value	A value explicitly present in the ground truth was not extracted in the corresponding field or record

Stage-specific performance and error attribution: To further identify the sources of errors throughout the extraction pipeline, validation was performed at the component level. The workflow was evaluated across PDF/table parsing and reconstruction, Task A experimental-condition extraction, Task B pitting-potential extraction, unit and reference-electrode representation, Task C condition–result alignment, and final database construction. Component-specific metrics were used according to the corresponding processing function. Precision, recall, and F1 score were calculated for Tasks A and B and for the final structured database, whereas exact accuracy was used for unit representation, pitting-potential normalization, reference-electrode representation, and condition–result alignment. Because independent intermediate ground-truth outputs were not retained for PDF/table parsing, preprocessing errors were assessed through retrospective error attribution rather than a standalone accuracy metric. Stage-specific error attribution was conducted at the component level to characterize where errors arose or became identifiable across the processing pipeline. The resulting stage-level counts represent component-level error occurrences rather than unique underlying extraction errors. Because a single underlying extraction error may manifest at, or be attributable to, more than one processing stage, the total number of stage-level error occurrences is not expected to equal the number of unique underlying errors classified by error type in Table 7. The resulting error-attribution percentages therefore describe the distribution of component-level error occurrences across the pipeline and should not be interpreted as component-specific error rates, because the number of eligible evaluation opportunities differed among components.

Treatment of missing outputs: To distinguish genuine source-level missingness from extraction failure, missing outputs were evaluated against the manually verified source during validation. When the corresponding information was not explicitly reported in the source, the missing field was treated as *source not reported* rather than as an extraction error. When the information was explicitly reported but was not captured or was incorrectly extracted by the pipeline, the case was classified as *not detected* or *wrong extraction*, respectively. Parsing- and alignment-related failures were separately identified through the stage-specific error-attribution analysis described above. This distinction ensured that source-level absence was not conflated with missingness or errors introduced by the extraction pipeline.

The unsupported-extraction rate was defined as the proportion of extracted values that could not be verified against the manually established ground truth. Values attributable to identifiable extraction errors, such as incorrect field assignment, unit errors, reference-electrode representation errors, table misalignment, or condition–result mismatch, were distinguished from unsupported extractions. The unsupported-extraction rate was calculated as:

$$R_{\text{unsupported}} = \frac{N_{\text{unsupported}}}{N_{\text{extracted}}} \tag{5}$$

The omission rate was defined as the proportion of explicitly reported eligible ground-truth values that were not captured by the extraction pipeline:

$$R_{\text{omission}} = \frac{N_{\text{missed}}}{N_{\text{eligible ground truth}}} \tag{6}$$

Evidence-grounding accuracy was defined as the proportion of extracted values for which the corresponding supporting evidence passage contained both the correct parameter and the correct numerical value. During execution of the framework, supporting evidence passages were generated at the relevant subtasks to guide extraction and condition–result association; however, these intermediate outputs were not persistently retained after completion of the full extraction workflow. Consequently, a retrospective field-level evidence-grounding accuracy could not be consistently calculated for the expanded validation set. Grounding reliability was therefore additionally assessed through manual comparison with the source-verified ground truth and the unsupported-extraction analysis described above.

Condition–result alignment accuracy was defined as the proportion of eligible condition–E_{pit} pairs in which the extracted pitting potential was associated with the correct alloy, specimen, and experimental condition. This metric was calculated for the 140 eligible condition–result pairs in the expanded validation set. Because the number of eligible cases was relatively limited, the resulting accuracy estimate was interpreted cautiously and was supplemented by the separate error-attribution analysis of incorrect condition–result associations.

Statistical analysis of the constructed database: For the stainless-steel-specific statistical analysis, material-domain screening was performed at the experimental-record level before complete-case selection. Records explicitly identifiable from the extracted material information as non-stainless materials, including carbon/low-alloy steels, aluminum alloys, Ni-based alloys, and other clearly non-stainless metallic materials, were excluded. A conservative exclusion strategy was adopted: records were excluded only when the available material information explicitly indicated a non-stainless material; otherwise, they were retained for subsequent analysis. Following material-domain screening and reference-electrode harmonization, complete-case selection was applied. Records were included only when the SCE-harmonized pitting potential and all variables included in the partial-correlation analysis contained valid numerical values; records containing missing values in any of these fields were excluded. This resulted in 1,379 records from 448 publications for the final partial-correlation analysis.

Individual experimental records were retained as the analytical units, while the dependence among multiple records originating from the same publication was accounted for using publication-clustered bootstrap resampling. In each of 2,000 bootstrap iterations, publications were sampled with replacement, and all records associated with each selected publication were retained as a cluster. Partial correlation coefficients were recalculated for each resampled dataset, and the 2.5th and 97.5th percentiles of the resulting distributions were used to obtain publication-clustered 95% confidence intervals. No additional observations were excluded solely on the basis of statistical outlier thresholds.

Reproducibility details: Complete prompts for Tasks A–C, exact model and API configurations, decoding settings, software versions, field definitions, normalization rules, and reference-electrode handling procedures are provided in the Supplementary Information.

Results and discussion

Comparative benchmark and model selection

To quantitatively determine the LLM configuration for large-scale database construction, five model configurations were comparatively evaluated using a representative validation subset of 100 publications. The same preprocessing procedures, task-decomposition workflow, prompt instructions, output schema, and evaluation criteria were applied to all configurations to ensure a controlled comparison. The benchmark results are summarized in Table 3.

As shown in Table 3, the proposed hybrid configuration achieved the highest overall extraction performance, with a precision of 91.30%, recall of 91.14%, and F1 score of 91.22%. Among the individual models, Qwen-Max showed the highest F1 score (88.43%), followed by DeepSeek-v3.2 (87.00%), Gemini Flash (84.81%), and GPT-3.5 (78.63%). Compared with Qwen-Max and DeepSeek-v3.2 used independently, the hybrid configuration improved the F1 score by 2.79 and 4.22% points, respectively. The improvement was also accompanied by a reduction in unsupported extractions, with the unsupported-extraction rate decreasing to 5.01%, compared with 6.72% for Qwen-Max and 7.83% for DeepSeek-v3.2.

The improved extraction performance of the hybrid configuration was achieved at the expense of longer processing time. Its average processing time was approximately 50 s per publication, compared with 28 s for Qwen-Max and 30 s for DeepSeek-v3.2. Nevertheless, the processing cost remained competitive at ¥0.05 per publication, which was lower than that of Qwen-Max (¥0.07), Gemini Flash (¥0.09), and GPT-3.5 (¥0.08), and only slightly higher than that of DeepSeek-v3.2 (¥0.04). These results indicate that the proposed hybrid configuration does not provide an advantage in inference speed but offers a favorable trade-off between extraction accuracy, hallucination control, and processing cost. Accordingly, the DeepSeek-v3.2–Qwen-Max hybrid configuration was selected for the subsequent large-scale database construction.

Condition–result alignment accuracy was not reported as a standalone quantitative metric because this metric is only applicable to publications containing multiple experimental conditions with independently distinguishable condition–E_{pit} pairs. Such cases were relatively limited in the 100-publication benchmark subset, and the resulting percentage would therefore be highly sensitive to a small number of individual mismatches. Instead, condition–result correspondence was incorporated into the record-level correctness assessment, whereby an extracted E_{pit} value was considered correct only when it was assigned to the corresponding experimental condition.

In addition to the performance metrics reported in Table 3, the operational reliability of the evaluated configurations was assessed in terms of structured-output validity and processing stability. Across all five model configurations evaluated on the 100-publication benchmark, the valid-JSON output rate was 100%, defined as the proportion of model responses that could be successfully parsed and conformed to the predefined JSON schema.

Table 3 Comparative performance of different LLM configurations.

	Precision	Recall	F1	Unsupported extraction	Time (s/publication)	Cost (¥/publication)
GPT-3.5	77.69%	79.60%	78.63%	14.39%	25	0.08
Gemini Flash	84.53%	85.68%	84.81%	9.46%	16	0.09
DeepSeek-v3.2	86.95%	87.05%	87.00%	7.83%	30	0.04
Qwen-Max	88.58%	88.29%	88.43%	6.72%	28	0.07
Hybrid	91.30%	91.14%	91.22%	5.01%	50	0.05

Table 4 Stage-specific performance of the extraction and database-construction pipeline.

Pipeline stage	Component	Metric	*N*	Performance (%)
I	PDF/table parsing	Error attribution	-	-
I	Task A: condition extraction	P/R/F1	31,500	94.64 / 93.79 / 94.21
II	Task B: E_{pit} extraction	P/R/F1	900	66.88 / 69.54 / 68.18
II	Unit representation	Accuracy	1195	94.39
II	E_{pit} normalization	Accuracy	140	66.43
II	Reference-electrode representation	Accuracy	194	90.21
III	Condition–result alignment	Accuracy	140	75.00
III	Final database	P/R/F1	38,700	93.83 / 93.26 / 93.54

Table 5 Distribution of identified errors across pipeline components.

Stage_ID	Stage	Attributed_Error_Count	Percentage_of_All_Errors
1	PDF/table parsing and reconstruction	25	2.13
2	Task A: Experimental-condition extraction	974	82.89
3	Task B: Pitting-potential extraction	23	1.96
4	Unit/reference-electrode representation	65	5.53
5	Task C: Condition–result alignment	88	7.49

No invalid JSON responses were observed during the benchmark. This high structured-output reliability is consistent with the constraint-guided zero-shot prompting strategy, in which the expected output schema and formatting requirements were explicitly specified. The processing failure rate was 0%, where processing failures were defined as unsuccessful extraction attempts caused by request timeout, context-length overflow, or incomplete responses resulting from token truncation. No such failures were observed across the 100 benchmark publications. This result suggests that the adopted input-length control mechanisms, including localized evidence retrieval and sparse output generation, were effective in maintaining stable processing within the evaluated dataset.

Stage-specific performance and error analysis

Stage-specific performance differed across the individual components of the extraction pipeline. Task A achieved a precision of 94.64%, recall of 93.79%, and F1 score of 94.21%, indicating reliable extraction of experimental conditions, material composition, and microstructural information. In comparison, Task B showed lower performance for pitting-potential extraction, with a precision of 66.88%, recall of 69.54%, and F1 score of 68.18%. For subsequent post-processing, the exact accuracies of unit representation, E_{pit} value/unit normalization, and reference-electrode representation were 94.39%, 66.43%, and 90.21%, respectively. Condition–result alignment achieved an accuracy of 75.00% among 140 eligible records. After integration of all extracted and processed fields, the final database achieved an aggregate precision of 93.83%, recall of 93.26%, and F1 score of 93.54%, indicating strong overall field-level extraction performance. However, this aggregate performance should not be interpreted as representing equivalent accuracy for each individual variable. In particular, the lower Task B F1 score, normalization accuracy, and condition–result alignment accuracy indicate that pitting-potential extraction and processing remain more challenging than the overall field-level extraction task. This distinction is particularly relevant because pitting potential serves as the outcome variable in the subsequent corrosion-statistical analysis. Because the evaluation units and eligible denominators differed among components, the stage-specific metrics were used to characterize the performance of individual processing functions rather than for direct numerical comparison across stages (see Table 4 and Table 5).

Error attribution further identified the primary sources of errors within the pipeline. Among the identified errors, 2.13% were attributed to PDF/table parsing and reconstruction, 82.89% to Task A experimental-condition extraction, 1.96% to Task B pitting-potential extraction, 5.53% to unit/reference-electrode representation, and 7.49% to condition–result alignment. The predominance of Task A in the absolute error count primarily reflects its substantially larger number of evaluated fields and extraction opportunities. Therefore, these percentages

represent the distribution of identified errors across pipeline components rather than component-specific error rates.

The source-level missingness audit showed that low database coverage did not necessarily indicate extraction failure. Dissolved oxygen was explicitly reported in only 89 of 900 evaluated opportunities (9.89%), of which 83 were correctly detected, whereas flow rate was reported in only 68 of 900 opportunities (7.56%), of which 62 were correctly detected. Similarly, microstructural descriptors were reported in 651 of 2,700 field-level opportunities (24.11%), with 597 correctly detected. These results indicate that the low coverage of these variables was predominantly associated with limited reporting in the source literature rather than failure of the extraction pipeline. Pitting potential showed a different pattern: it was reported in 140 of 900 evaluated records (15.56%), of which 99 were correctly extracted, 10 were not detected, and 31 were incorrectly extracted. Thus, the relatively low coverage of pitting-potential information reflects both source-level sparsity and greater extraction difficulty.

Database construction and characteristics

Using the proposed extraction framework, 36,095 structured experimental records were obtained from the broad candidate corpus of 12,792 publications. Table 6 summarizes the number of available entries and the corresponding coverage rate for each extracted field within the complete extraction output.

Table 6 The final extraction quantity and coverage rate of each field.

Field Name	Number of Records	Coverage Rate (%)	Variable Type
E_{pit}	3756	10.41%	Numerical
scan_rate	17,717	49.08%	Numerical
Chloride_Concentration	25,798	71.47%	Numerical
Temperature	25,645	71.05%	Numerical
pH	9630	26.68%	Numerical
Other_anion_concentration	8229	22.80%	mixed-type
Cations	6132	16.99%	mixed-type
dissolved_oxygen_value	2766	7.66%	Numerical
Flow_Rate	821	2.27%	Numerical
Pre_immersion_Time	8218	22.77%	mixed-type
comp_Fe_wt	20,704	57.36%	Numerical
comp_Cr_wt	17,436	48.31%	Numerical
comp_Ni_wt	16,772	46.47%	Numerical
comp_Mo_wt	10,912	30.23%	Numerical
comp_Mn_wt	15,944	44.17%	Numerical
comp_Si_wt	14,878	41.22%	Numerical
comp_C_wt	14,653	40.60%	Numerical
comp_N_wt	4877	13.51%	Numerical
comp_P_wt	8908	24.68%	Numerical
comp_S_wt	9032	25.02%	Numerical
comp_Cu_wt	7958	22.05%	Numerical
comp_Ti_wt	5700	15.79%	Numerical
comp_Nb_wt	2999	8.31%	Numerical
comp_La_wt	63	0.17%	Numerical
comp_Ce_wt	267	0.74%	Numerical
comp_Te_wt	22	0.06%	Numerical
comp_V_wt	2610	7.23%	Numerical
other_composition(name)	16,315	45.20%	mixed-type
matrix_phase	7837	21.71%	categorical
inclusion_description	1542	4.27%	mixed-type
precipitate_description	2134	5.91%	mixed-type
Heat_Treatment_Process	6053	16.77%	mixed-type

For the stainless-steel-specific descriptive analysis, records were screened according to material identity before the distributions of key environmental factors and alloying elements were analyzed. For the pitting-potential distribution, reference-electrode harmonization was additionally applied to improve comparability across studies. Records corresponding to non-stainless materials were excluded from these analyses. The resulting distributions are presented in Figs. 9, 10 and 11.

Overall, the database exhibits clear research-preference characteristics, indicating that existing studies are predominantly focused on room-temperature, neutral chloride-containing environments and conventional austenitic and duplex stainless steel systems, whereas investigations under extreme environmental conditions or involving specialized alloy compositions remain relatively limited. With respect to environmental parameters, the temperature distribution is highly concentrated within the range of 20–40 °C, accounting for the vast majority of the collected records. This observation suggests that most pitting corrosion studies have been conducted under room-temperature or near-room-temperature conditions, which is consistent with common practices in laboratory-based electrochemical testing and marine environment simulation studies. The pH distribution exhibits a similarly concentrated pattern, with more than 80% of the data falling within the pH range of 7–8. This finding indicates that the majority of studies in the database were performed in neutral or mildly alkaline chloride-containing environments, such as natural seawater, artificial seawater, and NaCl-based electrolyte systems. In contrast, the chloride ion concentration distribution displays a pronounced right-skewed pattern. Approximately two-thirds of the records are concentrated within the range of 0.35–0.70 M, corresponding to a 3.5 wt% NaCl solution (approximately 0.6 M). This result further confirms that marine corrosion environments constitute the primary application scenario in contemporary pitting corrosion research, whereas studies conducted under highly concentrated chloride environments represent only a relatively small proportion of the available data.

From the perspective of alloy composition, the chromium (Cr) content is predominantly concentrated within the range of 14–22 wt%, with a peak occurring between 14 and 18 wt%. This distribution is highly consistent with the chemical compositions of widely used corrosion-resistant stainless steels, including 304, 316, and duplex stainless steels, indicating that the material systems represented in the database are broadly representative of those commonly investigated in the literature. The nickel (Ni) content exhibits a substantially broader distribution, with

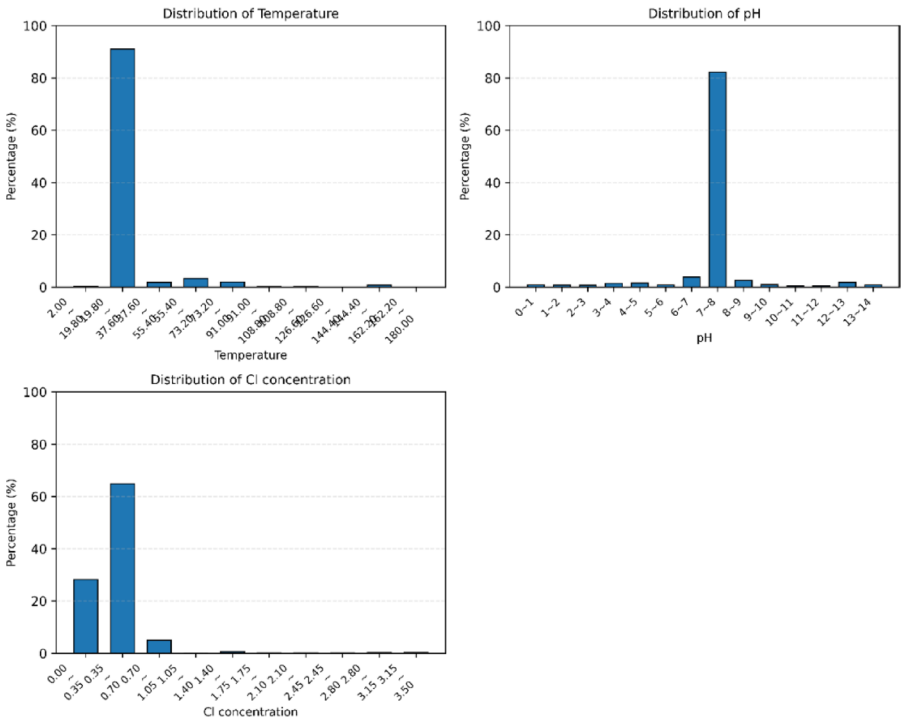


Fig. 9 Distribution of pH, temperature, and Cl concentration (non-missing records: $N = 1409$, 1493, and 1466, respectively).

https://doi.org/10.1038/s41598-026-75356-9

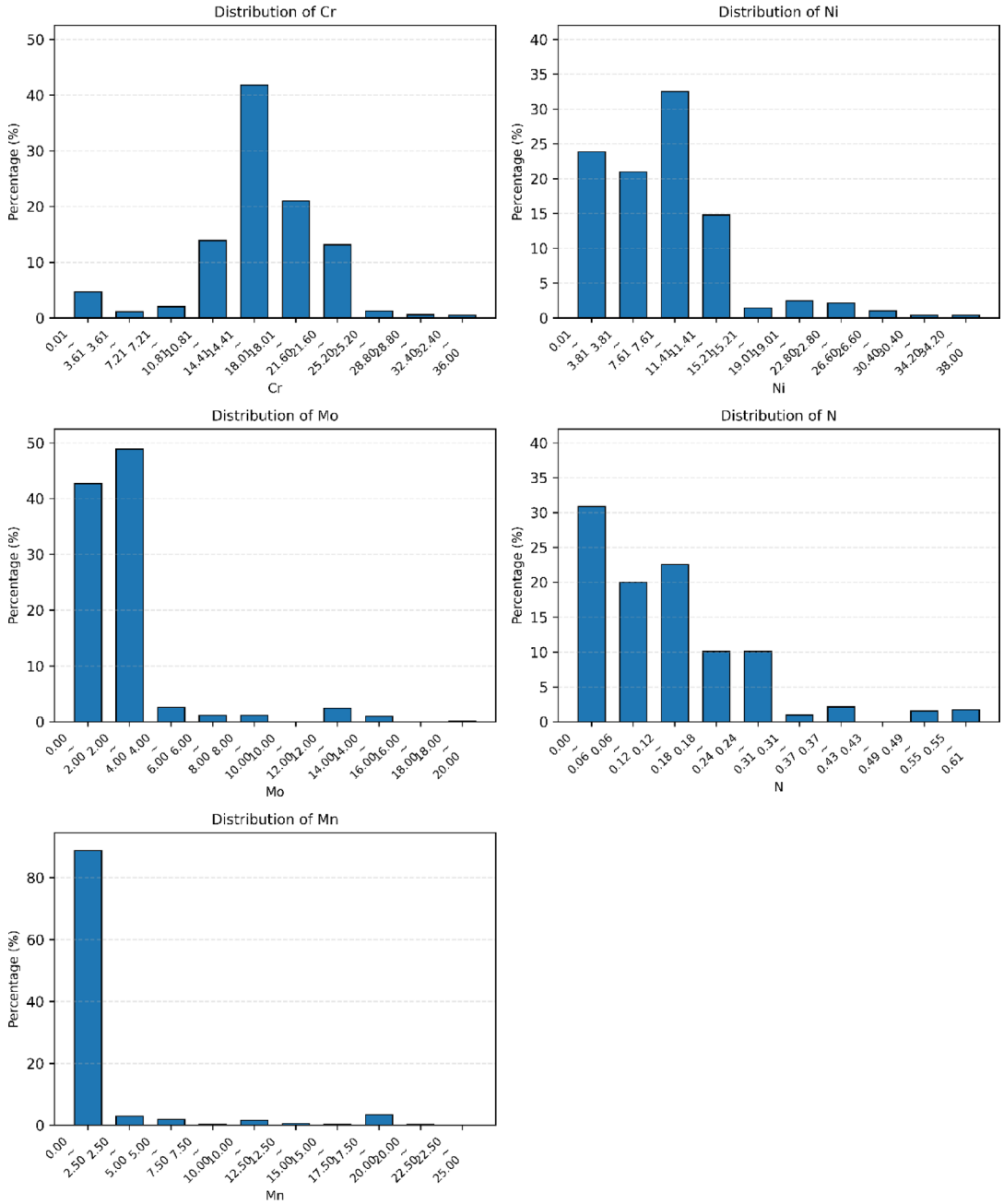


Fig. 10 Distribution of Cr, Ni, Mo, N, and Mn contents (non-missing records: $N = 1409$, 1258, 1035, 515, and 1232, respectively).

most records falling within the range of 0–12 wt%. This pattern reflects the inclusion of diverse stainless steel families within the database, including low-nickel ferritic stainless steels, high-nickel austenitic stainless steels, and various duplex stainless steels. In contrast, the molybdenum (Mo) content displays a pronounced right-skewed distribution, with the majority of samples concentrated between 0 and 4 wt%. The intervals of 0–2 wt% and 2–4 wt% account for the largest proportion of records, corresponding closely to the prevalence of 304 stainless steel (typically containing little or no Mo) and 316 stainless steel (typically containing approximately 2–3 wt% Mo), respectively.

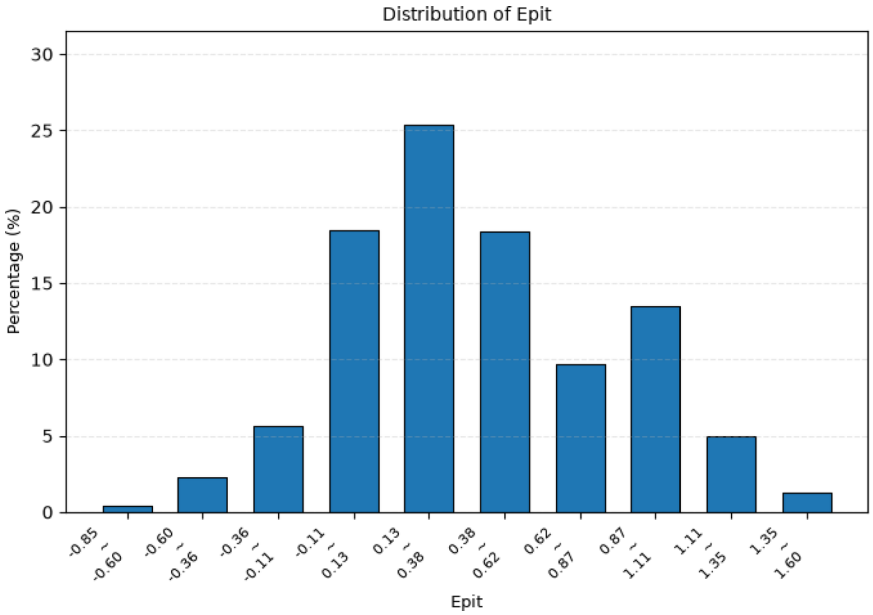


Fig. 11 Distribution of pitting potential ($N = 1409$ non-missing records).

For the minor alloying elements, the nitrogen (N) content is primarily distributed within the range of 0–0.3 wt%, while samples containing more than 0.4 wt% N are relatively scarce. This distribution is consistent with the typical nitrogen addition levels employed in austenitic and duplex stainless steels. The manganese (Mn) content exhibits a pronounced low-concentration clustering pattern, with the majority of records containing less than 2.5 wt% Mn. This observation is in agreement with the compositional characteristics of conventional stainless steel systems. Only a limited number of samples exhibit elevated Mn contents, which may correspond to high-manganese stainless steels or specialized alloy systems. Overall, the distributions of both N and Mn are consistent with established alloy design principles, and no obvious accumulation of anomalous values is observed within the database.

The distribution of pitting potential exhibits an approximately unimodal pattern, with the majority of records concentrated within the range of 0–0.6 V and a peak occurring between 0.1 and 0.4 V. This distribution is generally consistent with the typical pitting potential ranges reported in the literature for 304 stainless steel, 316 stainless steel, and duplex stainless steels. A limited number of records with negative pitting potentials likely correspond to materials with relatively poor corrosion resistance or to testing conditions involving more aggressive corrosive environments. In contrast, samples exhibiting E_{pit} values greater than 1 V are likely associated with high-PREN super duplex stainless steels or specialized corrosion-resistant alloy systems.

Overall, the distribution of E_{pit} spans a broad range of corrosion resistance levels, covering material systems from relatively low pitting resistance to highly corrosion-resistant alloys.

Collectively, the above results indicate that the experimental conditions and material systems represented in the constructed database are highly consistent with those that dominate contemporary research on stainless steel pitting corrosion. Overall, the database exhibits a characteristic distribution pattern centered on room-temperature, neutral chloride-containing environments and conventional austenitic and duplex stainless steel systems. At the same time, a proportion of records associated with less frequently represented environmental conditions and specialized alloy systems has also been retained, reflecting the diversity of research contexts represented in the literature corpus. These distribution characteristics not only reflect the prevailing research interests within the current literature but also provide a reliable data foundation for subsequent studies on pitting potential prediction, key influencing factor analysis, and data-driven materials design.

Validation of extraction performance

The extraction performance of the proposed framework was evaluated using the expanded manually validated dataset described in Sect. "Validation and statistical evaluation". Across the validation set, 10,828 true-positive (TP) values, 712 false-positive (FP) values, and 783 false-negative (FN) values were identified based on comparison with the manually established ground truth. These counts resulted in a precision of 93.83%, a recall of 93.26%, and an F1 score of 93.54%.

Publication-level bootstrap resampling was further used to quantify the uncertainty associated with these performance estimates. The resulting 95% confidence intervals were 93.14–94.49% for precision, 92.68–93.80% for recall, and 92.98–94.06% for the F1 score. The relatively narrow confidence intervals indicate limited uncertainty in the estimated extraction performance across the validated publications.

To further characterize extraction reliability beyond the aggregate precision, recall, and F1 score, all identified extraction errors were analyzed according to the eight predefined error categories described in Sect. "Validation and statistical evaluation". The resulting error distribution is summarized in Table 7.

A total of 1,125 underlying extraction errors were identified and classified according to their primary error mechanism. Hallucinated values constituted the largest error category, accounting for 596 cases (52.98% of all errors) and affecting 125 publications. Missed values represented the second-largest category, with 463 cases (41.16%) across 108 publications. Together, hallucinated and missed values accounted for 94.13% of all identified errors, indicating that the dominant failure modes were unsupported extraction and omission rather than formatting or unit-related errors.

Errors related to condition–result association were substantially less frequent, with 46 cases (4.09%) identified across 20 publications. Incorrect units, table row or column misalignment, and incorrect reference-electrode representation accounted for 1.07%, 0.62%, and 0.09% of the errors, respectively. No wrong-field assignments or duplicated records were identified in the validated sample.

In addition to the error composition, unsupported-extraction and omission rates were calculated separately to distinguish unsupported generation and complete omission from other extraction errors. Among 11,540 extracted values, 596 could not be verified against the manually established ground truth, corresponding to an unsupported-extraction rate of 5.16%. Among 11,611 eligible ground-truth values, 463 explicitly reported values were completely omitted by the pipeline and were therefore classified as "missed value" errors, corresponding to an omission rate of 3.99%. This omission count represents complete extraction failures and is therefore lower than the total number of FN events, which also includes ground-truth values that were not correctly matched because of other extraction errors. A standalone evidence-grounding accuracy was not retrospectively reported because the intermediate supporting evidence passages generated during extraction were not persistently retained after completion of the pipeline.

The observed reduction in recall can be attributed to several factors. First, to ensure the reliability of the constructed database, the prompting strategy employed stringent extraction constraints. When encountering ambiguous or insufficiently supported information, the model tended to adopt a conservative extraction behavior, prioritizing precision over recall. Consequently, potentially relevant data were occasionally omitted rather

Table 7 Distribution of extraction errors in the manual validation set.

Error	Count	% all errors	Publications
Hallucinated value	596	52.98%	125
Wrong field assignment	0	0	0
Incorrect unit	12	1.07%	11
Incorrect reference-electrode representation	1	0.09%	1
Table row or column misalignment	7	0.62%	6
Incorrect condition–result association	46	4.09%	20
Duplicated record	0	0	0
Missed value	463	41.16%	108

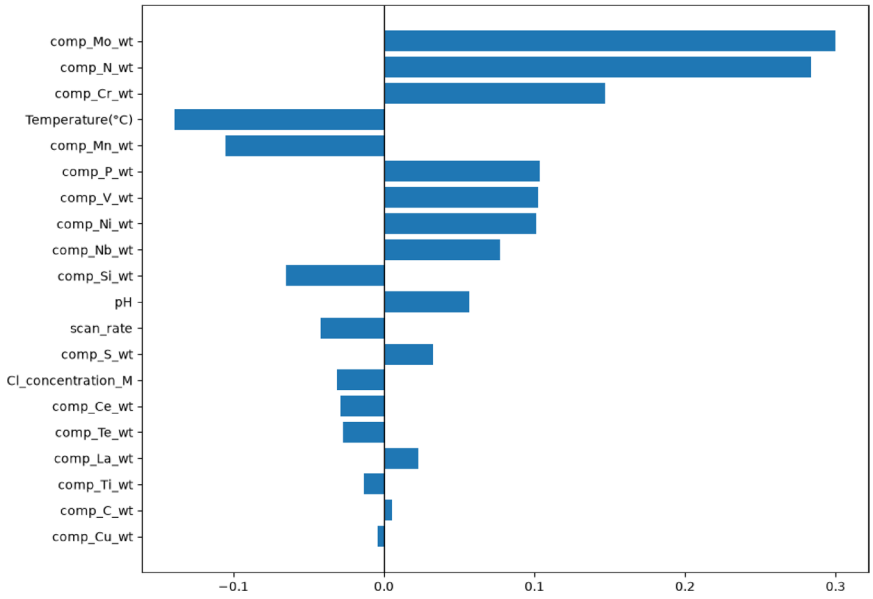


Fig. 12 Partial correlation analysis (N = 1,379 records from 448 publications).

than extracted with uncertainty. Second, experimental information is often dispersed across lengthy scientific documents, making it challenging for the model to establish long-range associations between related pieces of information. Third, limitations in the available token context may lead to attention dispersion, causing relevant information to be overlooked during the extraction process.

The extraction errors can likewise be attributed to several factors. First, numerical values located in close proximity to one another may be incorrectly associated with the target variable, resulting in erroneous value matching. Second, extraction errors may arise from failures in table reconstruction. In particular, when processing flattened PDF tables, the model may incorrectly associate values with the wrong columns or rows, leading to inaccurate data assignments. Third, although hallucination has been substantially reduced in modern Large Language Models (LLMs), it cannot be completely eliminated in large-scale literature processing. In rare cases, the model may generate outputs influenced by implicit assumptions, prior knowledge, or default reasoning patterns rather than strictly adhering to the source text.

Overall, the evaluation results demonstrate that the proposed LLM-based framework achieves strong aggregate performance in large-scale structured data extraction from scientific literature. However, the stage-specific evaluation also reveals heterogeneous performance across different components of the extraction pipeline, with pitting-potential extraction, normalization, and condition–result alignment remaining more challenging than the overall field-level extraction task. These results support the utility of the framework for automated database construction in corrosion informatics while also highlighting the need to account for field-specific extraction uncertainty when interpreting downstream analyses.

To clarify the material scope of the complete extraction output, the extracted records were additionally screened at the record level based on the available material information. Approximately 34% of the records were identifiable as stainless steels, 44% as non-stainless materials, while the remaining 22% could not be unambiguously classified based on the available information. Stainless-steel-specific downstream analyses were conducted using material-screened datasets, as described in Sect. "Validation and statistical evaluation".

Following material-domain screening, reference-electrode harmonization, and complete-case selection, 1,379 records from 448 publications were included in the final partial-correlation analysis. Partial correlation analysis was performed to evaluate the conditional associations between the investigated environmental parameters, alloying elements, and pitting potential. The results are presented in Fig. 12.

The partial correlation analysis revealed varying conditional associations between the investigated environmental factors, alloying elements, and pitting potential. Among the investigated variables, molybdenum (Mo),

nitrogen (N), and chromium (Cr), which are widely recognized as key pitting-resistance-enhancing elements, exhibited positive conditional associations with pitting potential. In contrast, temperature showed a negative conditional association. These statistical patterns characterize the relationships observed within the constructed cross-literature dataset and should be interpreted as conditional associations rather than causal effects.

Publication-clustered bootstrap analysis was further performed to account for dependence among records originating from the same publication. The positive conditional associations of Mo ($r=0.300$, 95% CI: 0.173–0.418), N ($r=0.284$, 95% CI: 0.191–0.362), Cr ($r=0.147$, 95% CI: 0.033–0.286), and Ni ($r=0.101$, 95% CI: 0.030–0.186) remained directionally robust, with their 95% confidence intervals excluding zero. Temperature ($r=-0.139$, 95% CI: −0.242 to −0.031) and Mn ($r=-0.105$, 95% CI: −0.185 to −0.017) showed negative conditional associations with confidence intervals excluding zero. In contrast, the confidence intervals of the remaining variables included zero. Chloride concentration showed only a weak negative partial correlation ($r=-0.031$), and its publication-clustered 95% confidence interval (−0.147 to 0.070) included zero.

Among all investigated variables, the molybdenum (Mo) content exhibited the highest positive partial correlation coefficient ($r=0.300$), representing the strongest positive conditional association with pitting potential. Molybdenum is widely recognized as a key alloying element for enhancing pitting corrosion resistance, primarily through improving passive film stability and increasing resistance to localized corrosion in chloride-containing environments. Its beneficial effects are generally attributed to the stabilization of Cr-rich passive films, the suppression of anodic dissolution within corrosion pits, and the promotion of repassivation processes[45,46].

Notably, the statistical association of Mo observed in this study is greater than that of nitrogen (N), which differs to some extent from the weighting assigned to N in the classical Pitting Resistance Equivalent Number (PREN) formulation, as shown in Eq. (7):

$$\mathrm{PREN} = \mathrm{Cr} + 3.3\mathrm{Mo} + 16\mathrm{N} \tag{7}$$

This discrepancy may arise from the fundamental difference between statistical relationships derived from large-scale literature data and the assumptions underlying idealized alloy design frameworks. The PREN equation is essentially an empirical formulation developed for specific stainless steel grades and representative chloride-containing environments, emphasizing the theoretical strengthening effects of alloying elements on pitting corrosion resistance. In contrast, the present study is based on a large collection of literature data spanning multiple stainless steel grades, experimental systems, and testing conditions. Consequently, the observed correlations reflect the overall statistical associations of alloying elements at a cross-study level rather than their idealized theoretical weighting. In addition, the database coverage of nitrogen (N) is relatively limited (13.51%), whereas molybdenum (Mo) exhibits substantially higher coverage (30.23%). As a result, high-nitrogen stainless steels constitute only a relatively small fraction of the overall dataset, which may partially reduce the apparent statistical association of N in the correlation analysis. Furthermore, the influence of N on pitting corrosion resistance is often strongly dependent on its metallurgical state, heat-treatment history, and nitride precipitation behavior. Consequently, its effect may vary considerably across different material systems and experimental conditions. In contrast, the beneficial effect of Mo on pitting corrosion resistance tends to be more consistent across a wide range of stainless steel systems. This greater stability may explain why Mo exhibits a stronger overall association in the large-scale cross-system statistical analysis performed in this study.

Nitrogen (N) nevertheless exhibits a strong positive partial correlation coefficient (0.284), indicating that it remains an important factor influencing the pitting corrosion resistance of stainless steels. Nitrogen is generally recognized for its ability to enhance austenite stability, promote chromium enrichment within the passive film, and improve local repassivation capability. Consequently, high-nitrogen stainless steels often exhibit elevated pitting potential values[47]. The results of this study suggest that, although the statistical association of N is lower than that of Mo in the large-scale literature-based analysis, it remains one of the most influential alloying elements for enhancing pitting corrosion resistance.

Chromium (Cr) exhibits a moderate positive partial correlation coefficient (0.147). As the principal alloying element responsible for the formation of the passive film, Cr plays a fundamental role in determining the pitting corrosion resistance of stainless steels[48]. However, because most stainless steels inherently contain relatively high Cr contents, the variation in Cr concentration across the database is comparatively limited. As a result, its statistical association in the partial correlation analysis is lower than those of Mo and N.

The observed effects of the major alloying elements are generally consistent with previous experimental studies on stainless steel pitting corrosion. In particular, the positive association of Mo with pitting potential agrees with its well-established role in enhancing passive-film stability and suppressing localized dissolution during pit initiation and propagation[46,49–50]. Similarly, the effects observed for Cr and N are broadly consistent with their established contributions to passivation and pitting resistance, which also form the basis of conventional pitting-resistance indices such as PREN[51]. However, the magnitudes of the associations observed in the present database are relatively modest compared with trends often reported under controlled experimental conditions. This difference is expected because the present analysis integrates heterogeneous stainless-steel grades, alloy compositions, solution chemistries, temperatures, and electrochemical testing conditions across a large number of independent studies. Therefore, the present results should be regarded as cross-literature statistical tendencies rather than direct substitutes for relationships derived from controlled experiments.

Among the environmental variables, temperature exhibits a clear negative partial correlation coefficient (−0.139), indicating that increasing temperature is associated with a reduction in pitting potential. This observation is consistent with established theories of pitting corrosion. Elevated temperatures can accelerate chloride ion (Cl^-) transport and anodic dissolution processes while reducing the stability of the passive film, thereby facilitating the initiation and stabilization of corrosion pits[52].

Manganese (Mn) also exhibits a negative partial correlation with pitting potential (−0.105). This observation may be closely associated with the presence of MnS inclusions. MnS inclusions are widely recognized as important initiation sites for pitting corrosion in stainless steels. Their preferential dissolution can promote local acidification and chloride ion (Cl^-) enrichment, thereby inducing localized destabilization of the passive film and facilitating pit initiation[53].

Notably, scan rate exhibited only a weak partial correlation with pitting potential ($r = -0.042$, 95% CI: −0.160 to 0.056), with the confidence interval spanning zero. From an electrochemical testing perspective, scan rate is generally considered an important experimental parameter that can influence the measured pitting potential. However, the present result indicates that no robust independent linear association between scan rate and pitting potential was evident in the heterogeneous cross-literature dataset. Higher scan rates may cause the polarization process to bypass certain stages of metastable pit development, thereby resulting in artificially elevated E_{pit} values. However, in the present large-scale cross-literature statistical analysis, the overall association of scan rate remains relatively limited. This observation may be attributed to the fact that scan rate primarily reflects methodological variations in experimental procedures rather than the intrinsic pitting corrosion resistance of the material itself. Furthermore, substantial differences often exist among studies with respect to electrode dimensions, potential scan ranges, IR compensation procedures, stabilization times, and test termination criteria. The combined influence of these factors may reduce the independent statistical association of scan rate within the overall dataset. Consequently, compared with experimental parameters, alloy composition and environmental conditions remain the dominant factors governing the pitting corrosion behavior of stainless steels.

In addition, chloride ion concentration exhibits only a weak negative partial correlation with pitting potential ($r = -0.031$, 95% CI: −0.147 to 0.070), with the confidence interval spanning zero. This result should not be interpreted as evidence that chloride ions (Cl^-) play an insignificant role in pitting corrosion. Rather, it suggests that pitting behavior is governed by the coupled effects of multiple interacting factors rather than by chloride concentration alone. Within the large-scale literature dataset, chloride concentration often varies simultaneously with other influential parameters, including temperature, pH, cation species, and buffering systems. Consequently, its independent association may be partially diminished in the partial correlation analysis. Furthermore, the influence of chloride concentration on pitting corrosion may exhibit a threshold-like behavior. Once a critical concentration

is exceeded, further increases in chloride concentration may produce a progressively smaller effect on the pitting potential. As a result, chloride concentration does not exhibit a particularly strong independent correlation in the cross-system statistical analysis conducted in this study.

It should be noted that the observed partial correlations were generally weak in magnitude, with the largest absolute coefficient being 0.300. Therefore, these results should not be interpreted as evidence of strong predictive or causal relationships. Rather, they indicate modest conditional associations within a heterogeneous cross-literature dataset involving diverse alloy systems and experimental conditions. The publication-clustered 95% confidence intervals further characterize the uncertainty associated with these estimates.

Overall, the partial correlation analysis results are broadly consistent with established theories of pitting corrosion in stainless steels, while also highlighting the differences between large-scale cross-literature statistical observations and idealized experimental systems. Although the observed associations are modest in magnitude, their general consistency with established corrosion knowledge provides a scientific plausibility check for the statistical patterns captured by the constructed database.

Conclusion

This study developed an LLM-based framework for structured data extraction and database construction from scientific literature and validated its effectiveness using stainless steel pitting corrosion research as a case study. By decomposing the complex information extraction process into multiple subtasks, including experimental condition extraction, pitting potential extraction, and data alignment, the proposed framework enabled the automated acquisition and integration of experimental data from large-scale unstructured scientific literature.

Using a broad candidate corpus of 12,792 publications, the framework generated a structured extraction output containing 36,095 experimental records, from which material-screened subsets were subsequently used for stainless-steel-specific analyses. The results demonstrate that the proposed framework is capable of effectively performing structured information extraction from complex scientific texts, with strong aggregate field-level performance but variable accuracy across individual extraction tasks. In particular, stage-specific validation identified pitting-potential extraction and normalization as comparatively challenging components of the workflow. These findings indicate that the framework provides a practical and scalable approach for high-throughput database construction in corrosion science, while emphasizing the importance of considering field-specific extraction uncertainty in downstream analyses.

Building upon the constructed database, statistical analyses were subsequently conducted. The results indicate that the material-screened stainless-steel dataset is predominantly characterized by room-temperature, neutral chloride-containing environments and austenitic and duplex stainless steel systems. Partial correlation analysis further revealed conditional associations between alloy composition, environmental factors, and pitting potential. Among the investigated variables, molybdenum (Mo) exhibited the strongest positive conditional association, whereas temperature showed a negative conditional association. In contrast, chloride ion concentration showed only a weak negative association. These statistical patterns are generally consistent with established corrosion knowledge, providing a scientific plausibility check while also highlighting differences between large-scale cross-literature statistical associations and traditional empirical models.

Overall, this study demonstrates the capability of LLMs for knowledge extraction and automated database construction from materials science literature and confirms their feasibility for applications in corrosion informatics. The proposed framework provides an effective pathway for transforming large volumes of unstructured scientific literature into structured, machine-readable data resources, thereby facilitating large-scale data analysis and knowledge discovery in corrosion science. The primary objective of the present study is database construction and validation, with the partial correlation analysis serving as an initial demonstration of the analytical utility of the resulting database. Future studies may further exploit this data resource using advanced data-driven approaches, such as clustering, machine learning, and other multivariate techniques, to identify complex

corrosion patterns and develop predictive models for pitting behavior. In addition, advances in multimodal LLMs, knowledge graphs, and materials ontology technologies may further enhance the capabilities of the proposed framework and enable the integration of diverse corrosion-related data sources, including polarization curves, electrochemical impedance spectroscopy (EIS) data, microstructural information, and surface morphology characterization results. These developments would support more comprehensive corrosion knowledge integration and data-driven discovery, providing new opportunities for materials design, performance prediction, and intelligent scientific research.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-026-75356-9.

Author contributions Z.H. conceived the study, designed the methodology, developed the large language model-based information extraction framework, performed data extraction, database construction, statistical analysis, and drafted the manuscript. J.D. contributed to methodology development, data analysis, and manuscript revision. Y.L. supervised the research, contributed to the study design, provided scientific guidance, and revised the manuscript. C.W. supervised the project, provided expertise in corrosion science, interpreted the results, and critically revised the manuscript. All authors reviewed and approved the final manuscript.

Data availability The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

1. Davis, J. R. (ed) *Stainless steels* (ASM international, 1994).
2. Di Schino, A. (ed) *Manufacturing and application of stainless steels* (MDPI, 2020).
3. Maziasz, P. J. & Busby, J. T. *Properties of austenitic stainless steels for nuclear reactor applications* (Oak Ridge National Laboratory (ORNL), 2011). Oak Ridge, TN (United States).
4. Frankel, G. S. Pitting corrosion of metals: a review of the critical factors. *J. Electrochem. Soc.***145** (6), 2186–2198 (1998).
5. Sedriks, A. J. *Corrosion of stainless steels* (Wiley, 1996).
6. Liu, M., Du, C., Liu, Z., Wang, L., Zhong, R., Cheng, X., … Li, X. (2023). A review on pitting corrosion and environmentally assisted cracking on duplex stainless steel.
7. Bhandari, J., Khan, F., Abbassi, R., Garaniya, V. & Ojeda, R. Modelling of pitting corrosion in marine and offshore steel structures–A technical review. *J. Loss Prev. Process Ind.***37**, 39–62 (2015).
8. Frankel, G. S. Pitting corrosion of metals: a review of the critical factors. *J. Electrochem. Soc.***145** (6), 2186–2198 (1998).
9. Chen, D. et al. Revealing the inner rules of PREN from electronic aspect by first-principles calculations. *Corros. Sci.***189**, 109561 (2021).
10. Soccol, D. An updated Pitting Resistance Equivalent Number by proportional hazard survival models of reported pitting potentials. *Electrochim. Acta*. **511**, 145355 (2025).
11. Li, B. et al. Studies on the cooperative influence of Cr and Mo on the pitting corrosion resistance of super austenitic stainless steels. *Materials***16** (23), 7397 (2023).
12. Frankel, G. S. Pitting corrosion of metals: a review of the critical factors. *J. Electrochem. Soc.***145** (6), 2186–2198 (1998).

13. Pardo, A. et al. Influence of pH and chloride concentration on the pitting and crevice corrosion behavior of high-alloy stainless steels. *Corrosion***56** (4), 411–418 (2000).
14. Dastgerdi, A. A., Brenna, A., Ormellese, M., Pedeferri, M. & Bolzoni, F. Experimental design to study the influence of temperature, pH, and chloride concentration on the pitting and crevice corrosion of UNS S30403 stainless steel. *Corros. Sci.***159**, 108160 (2019).
15. Tan, H. et al. Effect of annealing temperature on the pitting corrosion resistance of super duplex stainless steel UNS S32750. *Mater. Charact.***60** (9), 1049–1054 (2009).
16. Meng, G. et al. Effect of Cl−on the properties of the passive films formed on 316L stainless steel in acidic solution. *Journal Mater. Science Technology*. **30** (3), 253–258 (2014).
17. Dastgerdi, A. A., Brenna, A., Ormellese, M., Pedeferri, M. & Bolzoni, F. Experimental design to study the influence of temperature, pH, and chloride concentration on the pitting and crevice corrosion of UNS S30403 stainless steel. *Corros. Sci.***159**, 108160 (2019).
18. Ramprasad, R., Batra, R., Pilania, G., Mannodi-Kanakkithodi, A. & Kim, C. Machine learning in materials informatics: recent applications and prospects. *npj Comput. Mater.***3** (1), 54 (2017).
19. Hill, J. et al. Materials science with large-scale data and informatics: Unlocking new opportunities. *MRS Bull.***41** (5), 399–409 (2016).
20. Ramakrishna, S. et al. Materials informatics. *J. Intell. Manuf.***30** (6), 2307–2326 (2019).
21. Kononova, O. et al. Opportunities and challenges of text mining in materials research. *Iscience*, **24**(3). (2021).
22. Brown, T. et al. Language models are few-shot learners. *Adv. Neural. Inf. Process. Syst.***33**, 1877–1901 (2020).
23. Chowdhery, A. et al. Palm: Scaling language modeling with pathways. *J. Mach. Learn. Res.***24** (240), 1–113 (2023).
24. Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., … McGrew,B. (2023). Gpt-4 technical report. arXiv preprint arXiv:2303.08774.
25. Dagdelen, J. et al. Structured information extraction from scientific text with large language models. *Nat. Commun.***15** (1), 1418 (2024).
26. Foppiano, L., Lambard, G., Amagasa, T. & Ishii, M. Mining experimental data from materials science literature with large language models: an evaluation study. *Sci. Technol. Adv. Materials: Methods*. **4** (1), 2356506 (2024).
27. Gupta, S., Mahmood, A., Shetty, P., Adeboye, A. & Ramprasad, R. Data extraction from polymer literature using large language models. *Commun. Mater.***5** (1), 269 (2024).
28. Dreger, M., Malek, K. & Eikerling, M. Large language models for knowledge graph extraction from tables in materials science. *Digit. Discovery*. **4** (5), 1221–1231 (2025).
29. Schilling-Wilhelmi, M. et al. From text to insight: large language models for chemical data extraction. *Chem. Soc. Rev.***54** (3), 1125–1150 (2025).
30. Han, R. et al. An empirical study on information extraction using large language models. *arXiv preprint*arXiv:2305.14450. (2023).
31. Xu, D. et al. Large language models for generative information extraction: A survey. *Front. Comput. Sci.***18** (6), 186357 (2024).
32. Jiang, X. et al. Applications of natural language processing and large language models in materials discovery. *npj Comput. Mater.***11** (1), 79 (2025).
33. Brown, T. et al. Language models are few-shot learners. *Adv. Neural. Inf. Process. Syst.***33**, 1877–1901 (2020).
34. Kojima, T., Gu, S. S., Reid, M., Matsuo, Y. & Iwasawa, Y. Large language models are zero-shot reasoners. *Adv. Neural. Inf. Process. Syst.***35**, 22199–22213 (2022).
35. Schick, T. & Schütze, H. Exploiting cloze-questions for few-shot text classification and natural language inference. In *Proceedings of the 16th conference of the European chapter of the association for computational linguistics: main volume* (pp. 255–269). (2021), April.
36. Wei, J. et al. Chain-of-thought prompting elicits reasoning in large language models. *Adv. Neural. Inf. Process. Syst.***35**, 24824–24837 (2022).
37. Jain, S. & Wallace, B. C. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (pp. 3543–3556). (2019), June.
38. Liu, N. F. et al. Lost in the middle: How language models use long contexts. *Trans. association Comput. linguistics*. **12**, 157–173 (2024).
39. Khot, T. et al. Decomposed prompting: A modular approach for solving complex tasks. *arXiv preprint*arXiv:2210.02406. (2022).
40. Wang, L. et al. Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models. In *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers)* (pp. 2609–2634). (2023), July.
41. Chen, Y. C. et al. Enhancing responses from large language models with role-playing prompts:a comparative study on answering frequently asked questions about total knee arthroplasty. *BMC Med. Inf. Decis. Mak.***25** (1), 196 (2025).

42. Huang, L. et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Trans. Inform. Syst.***43** (2), 1–55 (2025).
43. Béchard, P. & Ayala, O. M. Reducing hallucination in structured outputs via Retrieval-Augmented Generation. *arXiv preprint*arXiv:2404.08189. (2024).
44. Liu, M. X. et al. we need structured output: Towards user-centered constraints on large language model output. In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (pp. 1–9). (2024), May.
45. Ha, H. Y., Lee, T. H., Bae, J. H. & Chun, D. W. Molybdenum effects on pitting corrosion resistance of FeCrMnMoNC austenitic stainless steels. *Metals***8** (8), 653 (2018).
46. Pardo, A. et al. Pitting corrosion behaviour of austenitic stainless steels–combining effects of Mn and Mo additions. *Corros. Sci.***50** (6), 1796–1806 (2008).
47. Sun, J., Tang, H., Wang, C., Han, Z. & Li, S. Effects of alloying elements and microstructure on stainless steel corrosion: a review. *Steel Res. Int.***93** (5), 2100450 (2022).
48. Yu, Y., Shironita, S., Souma, K. & Umeda, M. Effect of chromium content on the corrosion resistance of ferritic stainless steels in sulfuric acid solution. *Heliyon*, **4**(11). (2018).
49. Ilevbare, G. O. & Burstein, G. T. The role of alloyed molybdenum in the inhibition of pitting corrosion in stainless steels. *Corros. Sci.***43** (3), 485–513. https://doi.org/10.1016/S0010-938X(00)00086-X (2001).
50. Kaneko, M. & Isaacs, H. S. Effects of molybdenum on the pitting of ferritic- and austenitic-stainless steels in bromide and chloride solutions. *Corros. Sci.***44** (8), 1825–1834. https://doi.org/10.1016/S0010-938X(02)00003-3 (2002).
51. Jargelius-Pettersson, R. Electrochemical investigation of the influence of nitrogen alloying on pitting corrosion of austenitic stainless steels. *Corros. Sci.***41** (8), 1639–1664. https://doi.org/10.1016/S0010-938X(99)00013-X (1999).
52. Dastgerdi, A. A., Brenna, A., Ormellese, M., Pedeferri, M. & Bolzoni, F. Experimental design to study the influence of temperature, pH, and chloride concentration on the pitting and crevice corrosion of UNS S30403 stainless steel. *Corros. Sci.***159**, 108160 (2019).
53. Pardo, A. et al. Pitting corrosion behaviour of austenitic stainless steels–combining effects of Mn and Mo additions. *Corros. Sci.***50** (6), 1796–1806 (2008).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Zhen Huang[1] · Jinyan Duan[2] · Yupeng Liu[1] · Changgang Wang[3]

✉ Yupeng Liu
liuyupeng@hit.edu.cn

✉ Changgang Wang
cgwang@imr.ac.cn

[1] School of Economics and Management, Harbin Institute of Technology, 2, West Wenhua Rd., Weihai 264209, China

[2] School of Economics and Management, Harbin Institute of Technology, 13 Fayuan Street, Nangang District, Harbin 150001, China

[3] Institute of Metal Research, Chinese Academy of Sciences, Shenyang 110016, PR China